



UNIVERSITY OF SCIENCE
HO CHI MINH CITY

Privacy

Adapted from Quinn's Slides

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2025

Outline

- What is privacy?
- Harms and benefits of privacy
- Data mining
- Discussion

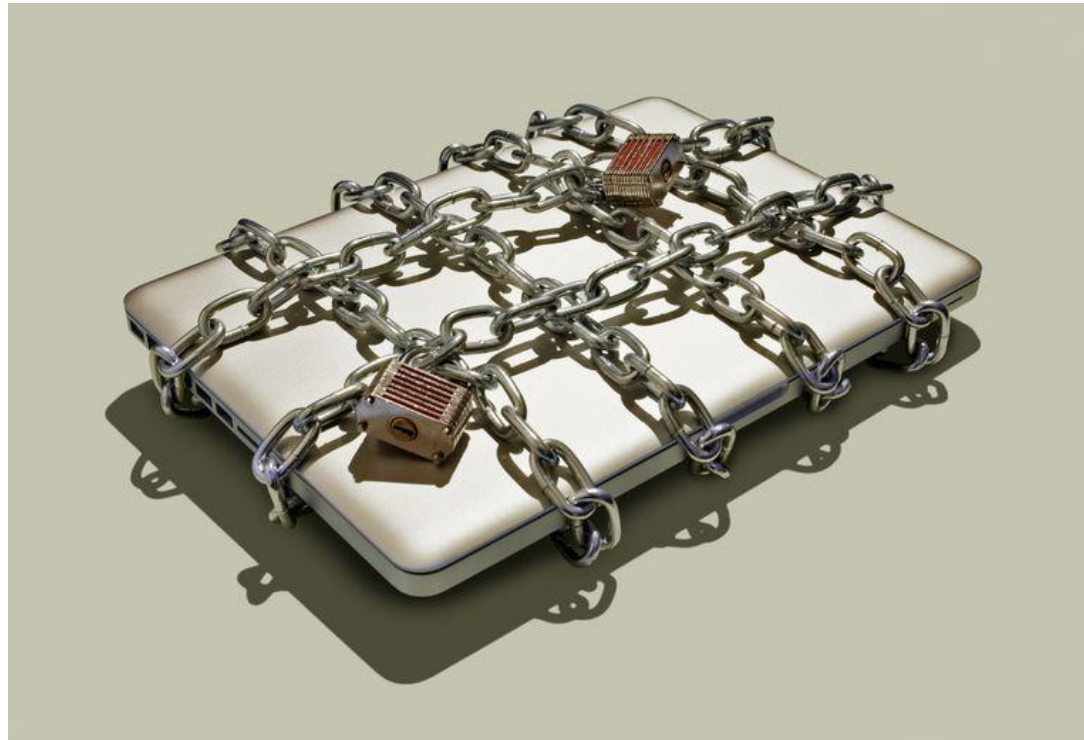


Image: lifewire.com

Số hóa > Công nghệ

Thứ ba, 12/7/2022, 16:18 (GMT+7)

Hacker rao bán 'dữ liệu trường học của 30 triệu người Việt'



Dữ liệu cá nhân kèm thông tin về trường lớp được cho là của 30 triệu người Việt đang bị tin tặc rao bán với giá 3.500 USD.

Bài rao được thành viên *meli0das* đăng trên một diễn đàn hacker từ ngày 8/7. Người này khẳng định thu thập được lượng dữ liệu lớn này "từ một website trường học phổ biến ở Việt Nam".

Hacker mô tả các thông tin bao gồm tên đăng nhập, email, số điện thoại, họ tên đầy đủ, ngày sinh, trường học và địa chỉ, là những dữ liệu chưa từng rò rỉ trước đây và mới được lấy trong tháng 7.

Để tăng sự tin tưởng, *meli0das* đăng ảnh chụp thông tin của khoảng 70 người, hầu hết là giáo viên. Ngoài ra, hacker này cũng cho biết có thể cung cấp bản xem trước của khoảng 10.000 người, hoặc có thể thương lượng để mua bán các gói dữ liệu nhỏ hơn.

uy8383	uyduong12@thaithuy.edu.vn	38082	hanh	hanh	Mon Sep 19 1983 07:00:00 GMT+0700 (+07)	TEACHER
hithaonguyenthpt	iaonguyen.thpt@gmail.com	31088	Thảo ...	Thảo Nguyễn	Fri Oct 06 1989 00:00:00 GMT+0700 (+07)	TEACHER
u7465	74c1xuankhe@gmail.com	30345	hị Hồng Kiên	hị Hồng Kiên	Sat Feb 16 1974 00:00:00 GMT+0700 (+07)	TEACHER
yet_lhp	tp@gmail.com	37890	h Nguyệt	h Nguyệt	Sun Apr 01 1984 07:00:00 GMT+0700 (+07)	TEACHER
kuin	@gmail.com	31525	Ca	Ca	Sun Aug 26 1979 00:00:00 GMT+0700 (+07)	TEACHER
thda1234	0408@gmail.com	38385	nh Thương	nh Thương	Mon Sep 02 1985 00:00:00 GMT+0700 (+07)	TEACHER
971gv1	@gmail.com	39569	HỊ THU HẢ	HỊ THU HẢ	Sat Mar 27 1971 00:00:00 GMT+0700 (+07)	TEACHER

What is privacy

- Privacy is a “zone of inaccessibility” (Edmund Byrne)
- Access
 - Physical proximity to a person, family, organization
 - Knowledge about a person, family, organization
- Examples of personal information
 - ID, date of birth, religion, love affairs



Privacy in the information age

- Personal data are easily exposed on the Internet or shared with services providers
 - Facebook/Google know almost everything from you
 - Your locations
 - What you like, don't like
 - What have you done and are going to do
 - Much more
- Should service providers use your data to make money?
- Concerns of privacy
 - SnapChat
 - Mark Zuckerberg had to cover his laptop's camera!

Privacy?

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Google loses 'right to be forgotten' case

🕒 13 April 2018

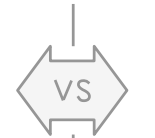
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A businessman fighting for the "right to be forgotten" has won a UK High Court action against Google.

Discussion

- What are harms and benefits of privacy?
- Everyone has the equal right to privacy?
 - Consider celebrities, public figures, government officials, common citizens

Harms and Benefits of Privacy

 Benefits		 Harms
 Individual growth		 Illegal activities
 Freedom		 Immoral activities
 Intellectual growth		 Burden on family
 Spiritual growth		 Dysfunctional families
 Trusting relationships		 Ignored fringes

Disclosing Information

Information categories



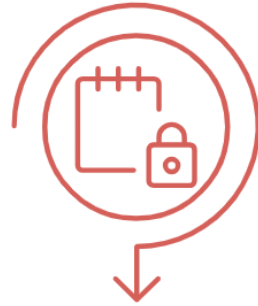
Public record

Information accessible to the public.



Public information

Information shared with an organization that can redistribute.

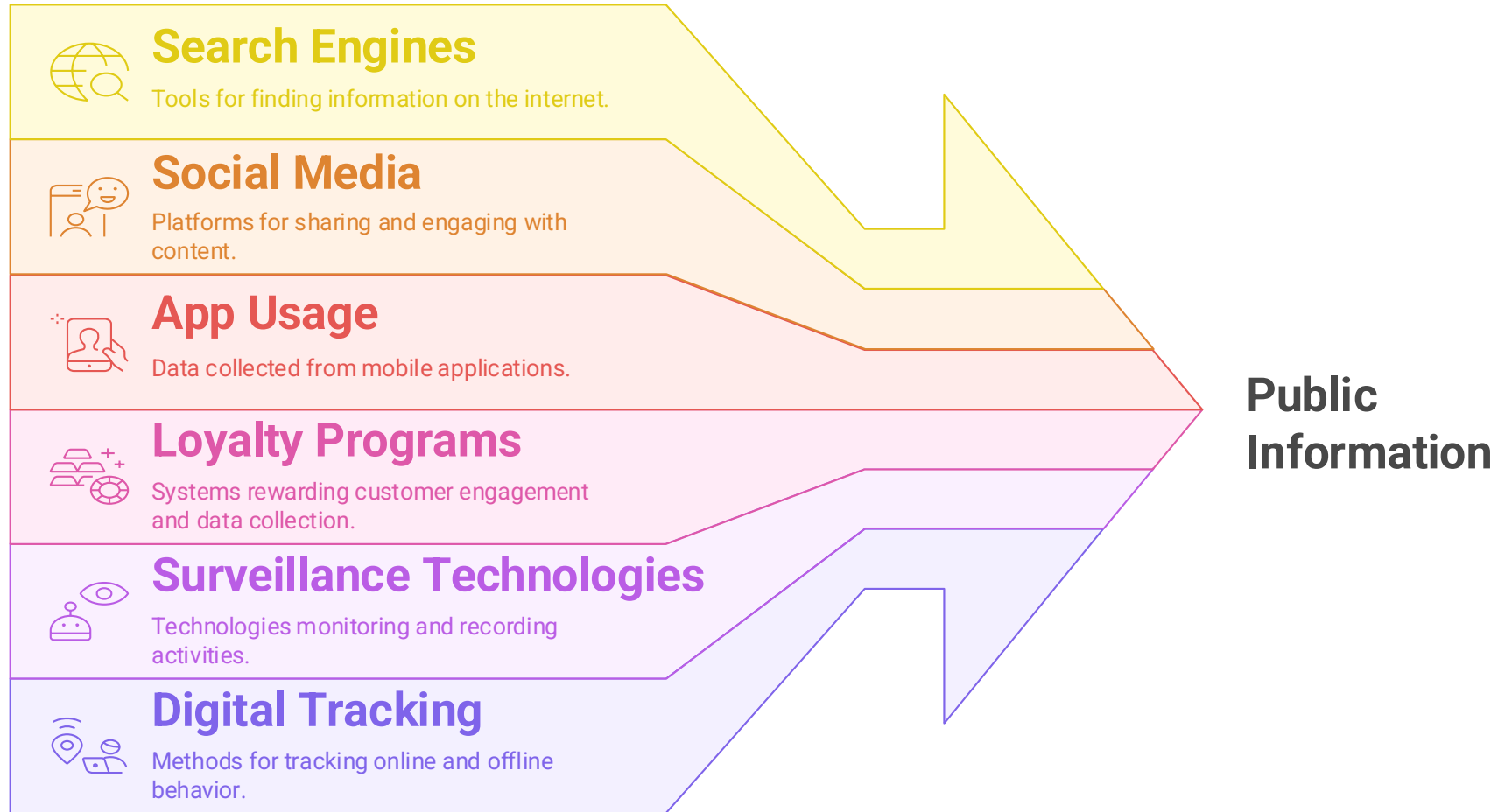


Personal information

Private information that is not disclosed.

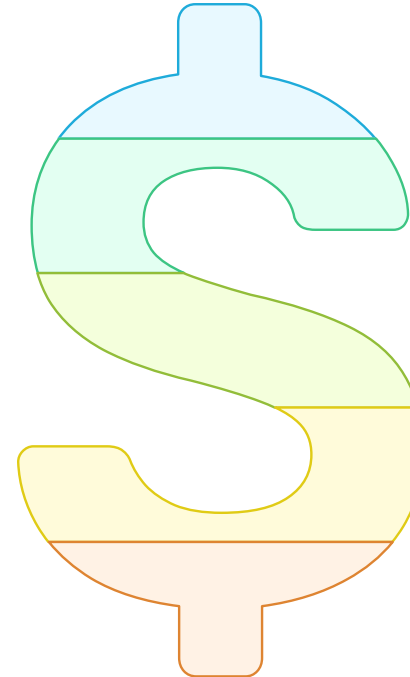
- Types of disclosures
 - ❑ Voluntary
 - ❑ Involuntary
 - ❑ Statutory

How information becomes public?



Data mining

- Data mining
 - Searching for patterns or relationships in one or more databases
 - Way to generate new information
- Information about customers is valuable



Amazon's Suggested Items

Personalized product recommendations based on user behavior



Netflix's Movie Suggestions

Tailored movie suggestions to enhance viewing experience



Google Search's Suggestions

Predictive search suggestions to improve search efficiency



Facebook's Fan Pages and Ads

Targeted ads and fan page suggestions to engage users



Targeted Ads

General trend of personalized advertising across platforms

Who owns transaction information?

- Buyers/consumers?
- Sellers/providers?
- Both?

- Does Facebook have the right to use your data?

- Opt-in
 - Consumers must explicitly give permission for orgs to share info
- Opt-out
 - Customers must explicitly forbid orgs from sharing info

Ways to protect privacy

- Encryption, using HTTPS for services that require personal information
- Using services with care
 - Do not share everything on Facebook, Zalo, etc.
- Never share personal information with service providers if you do not know them and how the data is used
- Using GPS carefully

AI Privacy Risks

Where AI systems put personal data at risk



Collection of Sensitive Data

Massive training datasets often sweep in health, biometric, and financial records, widening exposure.



Collection Without Consent

Personal data is often gathered for AI training without people's knowledge or agreement.



Use Without Permission

Data shared for one purpose, like a photo or resume, gets repurposed to train AI models.



Unchecked Surveillance & Bias

AI analysis of surveillance data can amplify bias, driving harmful outcomes like wrongful arrests.



Data Exfiltration

Attackers can manipulate AI, including via prompt injection, into stealing sensitive data.



Data Leakage

AI systems can accidentally expose one user's private information to another user.

Case study - Use of irrelevant and inaccurate personal information

- An agency employee wants to use a generative AI tool to assess applications for the agency's entrepreneurship grants. To do so, the employee uploads each individual's written application, as well as a spreadsheet containing the personal information of all applicants.
- In their prompt, the employee asks the Gen AI tool to summarise each application and compare the strength of the applications to identify preferred candidates.
- In preparing these summaries and recommendations, the Gen AI tool considers irrelevant matters and over-emphasises the strengths of certain demographic groups. Additionally, the system does not apply the selection criteria set out in the grants application process. The outputs of the system therefore contain irrelevant and inaccurate information.
- The agency employee relies on the recommendations of the Gen AI tool to approve the grants and does not manually validate the accuracy of its outputs. As a result, several highly qualified applicants are denied the grants, causing them to miss out on potential financial benefits. In addition, the personal information of all applicants has now been disclosed to the generative AI provider without their knowledge.

■ Source: <https://www.ipc.nsw.gov.au/resources/guide-privacy-risks-associated-use-generative-ai-tools>

Discussion 1

- Discuss the privacy concern of the bio data collection and use for bank transfer confirmation

Discussion 2

- Mr. Nguyen Duc Chung, former chairman of People's Committee of Hanoi proposed selling citizens' data to businesses. He estimated that the revenue for the city is 300 billion VND per year
- Are you for or against this proposal?
- Divide the class into two groups (for and against) to debate this issue

Discussion 3

- Recently many photos and articles concerning houses and assets of public officials have been shared on the Internet
- Discuss whether is it right or wrong to do so, considering different ethical theories (utilitarianism - harms and benefits, social contracts – laws and regulations, Kantianism)

Discussion 4

- The ministry of education and training (Moet) allows everyone to check grades of students taking the national high school examination using student exam id.
- Discuss whether is it right or wrong to do so